

Jifan Zhang

CS Ph.D. Candidate at University of Wisconsin-Madison

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Education

Sep 2021 - May 2026 University of Wisconsin, Madison

Ph.D. in Computer Science

Research focuses:

1. Data efficient learning of foundation models.

2. Humor abilities of multimodal LLMs.

Advisor: Robert Nowak

Sep 2016 - Mar 2021 University of Washington, Seattle

Mar 2021 M.S. in Computer Science, GPA: 3.98 / 4.0

Mar 2019 B.S. in Computer Science (minor in Mathematics), GPA: 3.94 / 4.0

Advisor: Kevin Jamieson

Publications

* Indicates equal contribution

Refereed

- **Humor in AI: Massive Scale Crowd-Sourced Preferences and Benchmarks for Cartoon Captioning**
Jifan Zhang*, Lalit Jain*, Yang Guo*, Jiayi Chen[†], Kuan Lok Zhou[†], Siddharth Suresh, Andrew Wagenmaker, Scott Sievert, Timothy Rogers, Kevin Jamieson, Robert Mankoff, Robert Nowak
Advances in Neural Information Processing Systems 37 (NeurIPS 2024). **Spotlight (top 3% submissions).**
- **AHA: Human-Assisted Out-of-Distribution Generalization and Detection**
Haoyue Bai, **Jifan Zhang**, Robert D Nowak
Advances in Neural Information Processing Systems 37 (NeurIPS 2024).
- **An Experimental Design Framework for Label-Efficient Supervised Finetuning of Large Language Models (Alphabetical)** Gantavya Bhatt*, Yifang Chen*, Arnav M Das*, **Jifan Zhang***, Sang T Truong, Stephen Mussmann, Yinglun Zhu, Jeffrey Bilmes, Simon S Du, Kevin Jamieson, Jordan T Ash, Robert D Nowak.
Findings of the Association for Computational Linguistics: ACL 2024.
- **LabelBench: A Comprehensive Framework for Benchmarking Adaptive Label-Efficient Learning**
Jifan Zhang*, Yifang Chen*, Gregory Canal, Arnav Das, Gantavya Bhatt, Stephen Mussmann, Yinglun Zhu, Simon Shaolei Du, Kevin Jamieson, Robert D Nowak.
Journal of Data-centric Machine Learning Research (2024). **Poster award at MMLS 2024.**
- **Learning from the Best: Active Learning for Wireless Communications**
Nasim Soltani*, **Jifan Zhang***, Batool Salehi, Debashri Roy, Robert Nowak, Kaushik Chowdhury.
IEEE Wireless Communications Magazine (2024).

- **Algorithm Selection for Deep Active Learning with Imbalanced Datasets**
Jifan Zhang, Shuai Shao, Saurabh Verma, Robert Nowak.
Advances in Neural Information Processing Systems 36 (NeurIPS 2023).
- **A Better Way to Decay: Proximal Gradient Training Algorithms for Neural Nets**
Liu Yang, Jifan Zhang, Joseph Shenouda, Dimitris Papailiopoulos, Kangwook Lee, Robert D. Nowak.
14th International OPT Workshop on Optimization for Machine Learning (at NeurIPS 2022).
- **GALAXY: Graph-based Active Learning at the Extreme**
Jifan Zhang, Julian Katz-Samuels, Robert Nowak.
Proceedings of the 39th International Conference on Machine Learning (ICML 2022).
Deployed in production at Meta through my internship. Internal shoutout from Mark Zuckerberg.
- **Improved Algorithms for Agnostic Pool-based Active Classification**
Julian Katz-Samuels, Jifan Zhang, Lalit Jain, Kevin Jamieson.
Proceedings of the 38th International Conference on Machine Learning (ICML 2021).

Preprint

- **SIEVE: General Purpose Data Filtering System Matching GPT-4o Accuracy at 1% the Cost**
Jifan Zhang, Robert Nowak (2024)
- **DIRECT: Deep Active Learning Under Imbalance and Label Noise**
Shyam Nuggehalli*, Jifan Zhang*, Lalit Jain, Robert Nowak (2024)
- **Learning to Actively Learn: A Robust Approach**
Jifan Zhang, Lalit Jain, Kevin Jamieson (2020)

Industry Experience

May 2022 - Dec 2022 - Meta, Core Data Science (CDS), Graph Science & Statistics Team
Research Intern / Part-time Student Researcher

- Built an Active Learning system from scratch handling **two million human annotations per month** on multi-modal data (language, image, video).
- Achieved 30% label-efficiency improvement in data diversity.
- The internship project (among two others) received **an internal shoutout from Mark Zuckerberg**.

May 2021 - Aug 2021 - Google, Ads pCTR Team
Software Engineering Intern

- Improved click-through-rate model **training efficiency by more than 10x** for efficient experimentation and hyperparameter search.
- Conducted trillion-scale data experiments on information theoretic data selection algorithms.

June 2020 - Sep 2020 - Google, Geo 3D Reconstruction Team
Software Engineering Intern

- Built next generation multi-view texturing algorithms for Google Map's 3D reconstructed models.
- Accelerated algorithm speed to accommodate web-scale rendering on all street view images.

June 2019 - Facebook, ML Data Platform Team

Sep 2019 *Software Engineering Intern*

- Prototyped distributed systems for large scale Machine Learning services of data pre-processing.
- Constantly communicated with and employed services from five different teams.

Research Experience

Sep 2021 - Wisconsin Institute of Discovery

Present *Research Assistant, advised by Professor Robert D. Nowak*

- Data-efficient learning and optimization in deep learning and/or with statistical guarantees.

Mar 2019 - Washington AI Lab

Mar 2021 *Research Assistant, advised by Professor Kevin Jamieson*

- Working on both theoretical and empirical perspectives of active learning algorithms
- Proposed novel learning to actively learn procedure for training optimal policy while obtaining same level of robustness as theoretically justified ones

June 2017 - Robotics and State Estimation Lab

June 2019 *Research Assistant, advised by Tanner Schmidt and Professor Dieter Fox*

- Built multi-tasking deep learning structures for computer vision and robotics tasks with both Back Propagation and Equilibrium Propagation (for energy based models)

June 2018 - SAMPL Group

Sep 2018 *Research Assistant, advised by Professor Zachery Tatlock*

- Worked on the Relay project (part of TVM, a deep learning library). Designing better intermediate tensor representation for the compiler

Teaching

Jan 2017 - University of Washington, Allen School of Computer Science and Engineering

Mar 2021 *Teaching Assistant*

- Graduate Courses: CSE 599G1 Deep Learning, CSE 546 Machine Learning
- Undergraduate Courses: CSE 446 Machine Learning, CSE 473(major)/415(non-major) Artificial Intelligence, CSE 341 Programming Languages
- Assist with overall course planning and development; design homework and exam problems; lead and prepare weekly sections; hold weekly office hours

Awards

- **Poster Award at MMLS 2024**
- **Microsoft Endowed Scholarship: recipient**
- **ACM-ICPC (Pacific Northwest Region): 5th place**
- **William Lowell Putnam Mathematical Competition: ranked 439 as sophomore and 733 as freshman**
- **UW Honors Calculus Award: One student per year awarded by the math department**